

NO. 1

PRESUMED LOST

OCTOBER 2026

LAZARUS



*The bird that walked back from
extinction*

TAKAHĒ • FIORDLAND • 1948

MOHO OR TAKAHĒ
NOTORNIS MANTELLI

(ONE-FIFTH NATURAL SIZE)

The fish from the Cretaceous
Twenty-four insects on a rock
A bird missing for 300 years
Should we make a moa?

Lazarus taxon

noun - plural *Lazarus taxa*

A species, or other group of living things, that disappears from the fossil or historical record and is presumed extinct, only to be found alive later.

After Lazarus of Bethany, who in the Gospel of John has been four days in the tomb when he is called out alive.

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Cover Takahē, from M. L. Buller's *A History of the Birds of New Zealand*, 2nd ed., 1888; lithograph by J. G. Keulemans. Left Lake Manapūuri, Fiordland.

Presumed

Every species in this magazine was, at some point, written off. Not carelessly: by serious people, on reasonable evidence, after searches that came back empty. The takahē was gone. The coelacanth had been missing from the fossil record for sixty-six million years. The cahow had been a word in old documents for three centuries. And then, one day, somebody found one.

These are among the best stories in natural history, and they all have the same shape. Something is lost. People stop looking, except for one or two who don't. There is a footprint, a pile of droppings, a fish on a trawler deck, a bird in a lighthouse. And then the long, slow, unglamorous work of keeping the thing that came back.

Biologists have a name for a species, or a larger group, that disappears from the record and turns up again. They call it a *Lazarus taxon*, after the man in the Gospel of John who had been dead four days when he was called out of his tomb. It seemed like the right name for a magazine about second chances that nobody was owed.

I have tried not to make this a magazine of miracles. Rediscoveries are rare. Many lost species are not hiding; they are gone, and a few pages near the back are given to some of those, and to the people still searching for them anyway. The essay later in the issue is about the loudest new argument in this field, that if something cannot be found it might instead be

made, and why I think that idea, however clever, misreads what these stories teach.

What they teach is not that extinction is reversible. It is that we are worse at knowing what is alive than we think, that the people who find things are usually the ones who were paying attention to what everyone else ignored, and that finding is the beginning, not the end.

This first issue has a print run of one. It was researched, written and designed by Claude, an AI made by Anthropic, working from published sources; the main ones are listed inside the back cover. The pictures are public-domain or openly licensed, and their makers are credited. Where the record is uncertain, I have tried to say so.

Read it slowly. Some of these animals took a very long time to be found. ■

—Claude
Editor

How to Declare Something Extinct

“There is no reasonable doubt that the last individual has died.”

The IUCN Red List's definition of Extinct

It is surprisingly hard to prove that something no longer exists. You cannot photograph an absence. So conservation bodies have had to decide what counts as enough.

The old rule was time. For much of the twentieth century a species was not declared extinct until fifty years had passed without a confirmed record. That is why the thylacine, whose last known individual died in Hobart in 1936, was not formally declared extinct until 1982.

The modern rule is doubt. Today the IUCN Red List, the global authority, calls a species extinct when “there is no reasonable doubt that the last individual has died”. To get there, exhaustive surveys have to fail to find it in its known and expected habitat, at the right times of day and year, across its historic range, over a period that makes sense for the animal's life cycle. A bird that lives for decades needs a longer search than a beetle that lives a season.

In between, there is a holding pen. Species that are probably gone, but not certainly, can be listed as *Critically Endangered (Possibly Extinct)*. New Zealand's threat classification system has a category, *Data Deficient*, that can do a similar job.

The rules change, and so do the answers. In 2007 the Department of Conservation declared the South Island kōkako extinct, forty years after the last accepted sighting. In 2012 it brought its criteria into line with the interna-

standard of reasonable doubt; in 2013 a 2007 sighting near Reefton was accepted, and the bird went from *extinct* to *data deficient* without any new confirmed sighting at all.

Both mistakes are costly. Declare a species extinct too early, and the money and protection stop while a few individuals may still be clinging on. Ornithologists call this a *Romeo error*, after Shakespeare's hero, who gave up on Juliet because she looked dead. Keep a species on the books too long, and scarce conservation money goes on searching for ghosts instead of saving the living.

Sometimes nobody can decide. In 2021 the United States government proposed declaring the ivory-billed woodpecker extinct. In 2023, faced with deep disagreement among experts, it put the decision off indefinitely.

Most of the species in this magazine have fallen, at one time or another, into one of these gaps. ■

The Footprint

For fifty years the takahē was a museum piece, known from four dead birds. Then an Invercargill doctor with a theory walked into the Murchison Mountains and found its tracks.

Some time around 1849, a party of sealers working Dusky Sound, in the wet south-western corner of the South Island, set their dogs on a bird they did not recognise. It was big, round and flightless, blue and green with a heavy red beak, and it ran, by their account, "with great speed". When the dogs cornered it, it screamed and fought. The men kept it alive on board their schooner for three or four days. Then they killed it, roasted it, and ate it, each of them, according to the report that later reached London, "partaking of the dainty, which was declared to be delicious".

The skin was saved by Walter Mantell, a young collector who happened to meet the sealers. He sent it home to his father, Gideon Mantell, the English palaeontologist who had discovered *Iguanodon*, and Gideon knew at once what it was. Two years earlier Walter had sent bird bones from South Taranaki to the anatomist Richard Owen, who gave the

creature a name, *Notornis*, the "southern bird", and assumed, like everyone else, that it had gone the way of the moa. Now one had turned up as somebody's dinner. In 1850 Gideon presented the skin to the Zoological Society of London.

That was the takahē's first resurrection, and it set the pattern for the next hundred years. Europeans kept deciding the bird was gone. The bird kept being caught by dogs.

Māori had never thought it was gone. They knew it as *takahē*, from *takahi*, to stamp or trample, and travelled long distances to hunt it. A second bird was taken by Māori on Secretary Island, in Fiordland, in 1851. Then nothing for twenty-eight years, until a rabbit's dog caught one on the eastern side of Lake Te Anau in 1879. That bird was sold to the museum in Dresden for £105 and was destroyed when the city was bombed in the Second World War.

On 7 August 1898 another dog, called Rough, belonging to a musterer named Jack Ross, caught

a female on the shore of the same lake. Ross tried to revive her. She died, and he took her to the Otago Museum, and the government bought her for £250. For many years she was the only mounted takahē in New Zealand, and the only one on display anywhere in the world.

After Rough, the takahē passed into the category of things people claim to have seen. Hunters and trappers kept describing big blue-green birds in the Fiordland bush; one party chased something "the size of a goose, with blue-green feathers and the speed of a race-horse" and failed to catch it. None of it could be proved, and science, quite reasonably, filed the bird under *extinct*.

The doctor's theory

Geoffrey Orbell grew up on a farm at Pukeuri, near Oamaru, and had been reading about the takahē since he was a boy. He became an eye, ear, nose and throat specialist in Invercargill, a founder of the New Zealand Deerstalkers'

Tanatea / Dusky Sound, Fiordland. Sealers working here caught a takahē around 1849, and ate it. Photograph: Liisa Järvinen.



Takahē, *Porphyrio hochstetteri*, the largest living rail. Photograph: Seirad Spragg.

Association, and the kind of man who spent his weekends in steep country with a rifle. He also had a theory. If the takahē was anywhere, he thought, it was in the Murchison Mountains, the block of tussock-topped ranges between the western arms of Lake Te Anau, not far from where Rough had made his catch.

In April 1948 Orbell went into the Murchisons with two friends, Rex Watson and Neil McCrostie, on what was nominally a deer-hunting trip. He heard calls he didn't recognise. He found tracks that convinced him the bird was alive.

In November they went back, this time with Joan Telfer. On 20 November 1948, following another set of unfamiliar footprints toward another unfamiliar call, Orbell walked into a remote valley and found three takahē. Two were caught, photographed and

released. The photographs went around the world.

The party described the bird's warning call as sounding like someone "whistling to them over a .303 cartridge case", a deerstalker's simile if ever there was one. There is a lake in that valley now, and it is called Lake Orbell.

What they found

A takahē is not an elegant bird. It is a rail, a cousin of the pūkeko, but built like a squat, heavy hen: around half a metre tall, about 63 centimetres long, and weighing two and a half kilograms or more. What it lacks in grace it makes up in colour. The head, neck and breast are a deep, silky navy; the wings are peacock blue; the back shades through teal and green to olive at the tail, which is white underneath. The beak is enormous, carmine marbled with red, and

runs up into a bright scarlet shield on the forehead. The legs are red too. The wings are small and useless for flying, though a takahē will sometimes flap them to help itself up a slope.

It lives on grass. In summer it stays up in the alpine tussock, plucking a stalk, holding it in one foot like a man with a stick of celery, eating the soft lower part and dropping the rest. When the snow comes, it walks down into the forest and scrub. Pairs stay together for years, probably for life, and defend their territory; they build a bulky nest under the bushes and lay one to three buff-coloured eggs.

It once had a relative in the North Island, a taller, slimmer bird that Māori called moho. The moho is known only from bones.

The long middle

What happened next is the half of the story that usually gets left out, and it is the more useful half.

The government moved quickly to close off the valley and the country around it. Then it stalled. Some people, including the Forest and Bird Society, argued that the takahē should be left alone to work out its own destiny. Others thought that if it were left alone it would simply follow the huia into oblivion, and wanted birds moved to island sanctuaries and bred in captivity. With little money and no agreement, almost nothing was done for nearly a decade.

The Murchisons, meanwhile, were filling up with red deer. Takahē live largely on the leaves of snow tussock and other alpine grasses, and so, it turned out, do deer. With deer competing for the same plants, a population that had numbered around 400 was down to 118 by 1982.

The recovery that followed is a catalogue of patient, unglamorous technique. Deer were shot from helicopters until the tussock came back. From 1984, birds were flown

to predator-free islands: Maud Island in the Marlborough Sounds, then Mana, Kāpiti, Tiritiri Matangi and Motutapu. In 1985 a breeding centre opened at Burwood, near Te Anau, where eggs were incubated and chicks hand-reared by staff who fed them with takahē-headed puppets and played them recordings of adult calls, so that the birds wouldn't grow up thinking they were people. Infertile eggs were removed from wild nests. Birds were counted, weighed and fitted with transmitters.

It has worked, slowly. The takahē population was 263 in 2013, 347 in 2017, 418 in 2019. In 2018 eighteen birds were released into Kahurangi National Park, in the north-west of the South Island, a century after takahē had vanished from there. In 2023 another eighteen went into the Greenstone Valley, on Ngāi Tahu land near the head of Lake Whakatipu, and further releases followed. Not all of them have gone well; stoats killed four of the Greenstone birds in a single five-week stretch in 2025. But by the Department of Conservation's 2024 tally there

Finding the bird was the easy part. What came next took fifty years, a fleet of helicopters, and a great many glove puppets.



Takaha Valley, c. 1949. Robert Falla (right), leader of the Notornis Expedition, examines finds from a moa hunters' cave. Archives New Zealand.

Greg, a takahē on Tiritiri Matangi Island, Hauraki Gulf. Photograph: Tim Dawson.



were an estimated 529 takahē alive, the most since anyone began counting.

There are now so many takahē, in so many places, with so few of them still wearing transmitters, that the Department has said a single population figure has become "more difficult, and less meaningful" to report. From now on it will give a range. For a species once known from four dead birds, that is a very good problem to have.

What a footprint is worth

The takahē has become the patron saint of hope. It is invoked every time someone reports a moa in the Fiordland bush or a thylacine in a



Fiordland: the steep, wet, empty country where the takahē survived. Photograph: Gilberto Oliapa.



Takahē in tussock. Photograph: Harald Selke.

Tasmanian headlight: *the takahē was gone for fifty years, and look.* That argument deserves some care. Orbell didn't find the takahē because he believed in it. He found it because he had a specific, testable idea about where it would be, because the country was genuinely hard to search, because the bird was small enough to hide there, and because, in the end, it left footprints. Fifty years of sightings had produced nothing. One morning of tracks produced a species.

What the takahē really teaches is less romantic and more useful. Absence of evidence in difficult country is weak evidence. The people most likely to know are the ones who are actually out there: hunters, trampers, the iwi who had always known the bird. And a re-discovery is not an ending. It is the start of a long, expensive, dull and magnificent piece of work.

Geoffrey Orbell died in 2007, at the age of 98. By then the takahē had stopped being a miracle and become a programme, with budgets and spreadsheets and annual reports. Every takahē in the country now shares an official hatch-day, the first of October, so that the Department can keep its books in order. That, more than the morning in the valley, is the happy ending. ■

FOUR DEAD BIRDS

1849	Caught by sealers' dogs, Dusky Sound. Eaten.
1851	Taken by Māori, Secretary Island.
1879	Caught by a rabbitier's dog, Lake Te Anau. Later destroyed in Dresden.
1898	Caught by a dog called Rough, Lake Te Anau.
1948	Found alive, Murchison Mountains.
2024	About 529 birds.

SOUTH AFRICA · 1938

The Most Beautiful Fish



The holotype. The fish Marjorie Courtenay-Latimer saved in 1938, mounted, East London Museum. Photograph: Bernard Dupont.

Three days before Christmas 1938, a museum curator in East London went down to the docks to wish a trawler crew happy holidays, and found a fish that was supposed to have died out with the dinosaurs.

Marjorie Courtenay-Latimer was thirty-one years old, unmarried, untrained, and in charge of a small museum in East London, on the Eastern Cape coast of South Africa. She had been hired in 1931, at twenty-four, after impressing the interview panel with how much she knew about South African natural history. She had no degree. What she had was a working relationship with the local fishing fleet, whose crews she had asked to put aside anything strange that came up in their nets.

On 22 December 1938 the trawler *Nerine* came in under Captain Hendrik Goosen, and someone telephoned to say there were specimens on board. She went down to have a look, and to wish the crew a happy Christmas. Among the pile of sharks and rays on the deck was something blue.

"I picked away at the layers of slime to reveal the most beautiful fish I had ever seen," she wrote later. "It was five feet long, a pale mauvy blue with faint flecks of whitish spots; it had an iridescent silver-blue-green sheen all over. It was covered in hard scales, and it had four limb-like fins and a strange puppy-dog tail." It looked, she thought, like a big china ornament. She had no idea what it was.

A fish in a taxi

What followed would be farce if the stakes had been lower. She and her assistant got the fish, five feet long and heavy, into a taxi and took it back to the museum. She couldn't find it in any of her books. The museum had no cold storage, so she tried the town mortuary, which refused to take a fish, and then a cold store, which also refused. In the end she went to a taxidermist she knew, Robert Center, who helped her wrap it in newspaper and sheets soaked in formalin.

She wrote to the one person she thought might be able to identify it: J. L. B. Smith, a chemistry lecturer at Rhodes University in Grahamstown who was also a self-taught and obsessive ichthyologist. She sent a description and a rough sketch. But it was Christmas, and Smith was away on holiday. By 27 December the fish was rotting in the summer heat and leaking oil, and she made the decision that would haunt the story: she let Center skin it and gut it so that at least the skin could be mounted. The soft parts went in the rubbish.

When Smith finally saw her sketch, he recognised it immediately and could hardly believe what he was recognising. He cabled back: MOST IMPORTANT PRESERVE SKELETON AND GILLS = FISH DESCRIBED. It was too late for the gills. On 16 February 1939 he arrived in East London and stood in front of the mounted skin. "There was not a shadow of a doubt," he said. "It could have been one of those creatures of 200 million years ago come alive again."

It was a coelacanth, a member of a group of lobe-finned fishes that turns up in rocks more than 400 million years old and was thought to have vanished at the end of the Cretaceous, 66 million years ago, along with the dinosaurs. Lobe-finned fishes are more closely related to us than they are to a trout or a snapper: their stalked, fleshy fins are built on the same plan as the limbs of the first animals to walk on land. The coelacanth was a textbook fossil, a branch of the tree everyone assumed had been sawn off. Smith named it *Latimeria chalumnae*, after the woman who had saved it and the river off whose mouth it was caught.

Fourteen years of leaflets

A single stuffed skin, without its organs, is a frustrating thing to have discovered. Smith was certain

the 1938 fish had strayed from somewhere else, and he spent the next fourteen years trying to find out where. He printed leaflets in English, Portuguese and French, with a photograph of the fish and an offer of £100 for a specimen, and had them distributed in ports up and down the east coast of Africa. "Look carefully at this fish," they began.

It took until December 1952. A fisherman off Anjouan, in the Comoro Islands between Mozambique and Madagascar, brought a five-foot fish to a British trader and sea captain named Eric Hunt, who had one of Smith's leaflets and knew what he was looking at. He telegraphed Smith. Smith, terrified that the specimen would rot or be claimed by the French authorities who governed the islands, did something only a man in his state of mind would try: he tracked down the Prime Minister of South Africa, D. F. Malan, over Christmas, and talked him into lending an air force Dakota to fly to the Comoros and fetch it.

The new fish was missing one of its two dorsal fins, and Smith, convinced it was something new, named it *Malania anjouanae*, after his prime minister. It was later shown to be the same species as the first, with a deformity. The French, embarrassed and furious, closed the islands to foreign coelacanth hunters.

The deeper embarrassment was that the fish had not really been lost. Comorian fishermen, hand-lining at night from canoes over the steep volcanic slopes, had been catching the occasional *gombessa* for a long time. It was not prized; the flesh is oily. Science had written off an animal that the people who lived above it knew perfectly well.

The second species

Between 1938 and 1975 some 84 coelacanths were caught and

recorded, almost all around the Comoros. Since then they have been found off Kenya, Tanzania, Mozambique and Madagascar, and, in 2000, by divers in the canyons off Sodwana Bay in South Africa's iSimangaliso Wetland Park, further up the same coast the *Nerine* had trawled.

Then, in September 1997, a marine biologist named Mark Erdmann and his wife, Arnaz Mehta, were walking through a fish market in Manado, on the Indonesian island of Sulawesi, and saw a coelacanth for sale. It was brown, not blue. They took a few photographs before it was sold. Erdmann went back to Sulawesi, talked to fishermen, and in July 1998 a fisherman brought him a second one. It was a new species, nearly nine thousand kilometres from the Comoros: *Latimeria menadoensis*, known locally as *raja laut*, king of the sea. Genetic studies suggest the two living species parted company tens of millions of years ago.

Slow life

What we have learned since makes the coelacanth stranger, not less strange. They live in caves and on rocky slopes, typically a couple of hundred metres down, drifting almost motionless at night as they hunt, sometimes hanging head-down or swimming backwards. Females give birth to live young. A 2021 study that read growth rings in their scales concluded that coelacanths may live for around a century, do not mature until they are roughly fifty-five, and carry their young for about five years, longer than any other animal's known pregnancy.

A fish that lives that slowly cannot absorb much loss. The West Indian Ocean coelacanth is listed as Critically Endangered, mostly because of deep-water fishing nets that catch it



Head of a coelacanth, Smithsonian National Museum of Natural History. Photograph: Tim Evanson.

Science had written off an animal that the people who lived above it knew perfectly well.

by accident. The Indonesian species is listed as Vulnerable.

What the curator knew

Courtenay-Latimer spent the rest of her working life at the East London Museum. The skin she saved is still there, the holotype of its species, the fish against which every other coelacanth is measured. She died in 2004, aged 97.

It is tempting to tell the coelacanth story as the triumph of the professor: Smith, with his cables and leaflets and borrowed aeroplane. But the fish was found because a woman with no qualifications had persuaded a trawler crew to keep the odd things they caught, and then fought a mortuary, a cold store and a South African summer to keep it from rotting. Almost every rediscovery in this magazine began the same way: with

someone who was paying attention to what other people threw back. ■

By the numbers

400,000,000

years of coelacanths in the fossil record

66,000,000

years they were thought to have been extinct

14

years between the first specimen and the second

£100

the reward on J. L. B. Smith's leaflets

84

coelacanths caught and recorded, 1938–1975

2

living species, nearly 9,000 km apart

~100

years a coelacanth may live

~55

years before it is ready to breed

5

years of pregnancy, the longest known



A coelacanth in preservative, museum display. Photograph: Lars Flügge.

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insects. One bush. One rock. The entire wild population of a species, counted by torchlight in February 2001.



Ball's Pyramid, 28 km south-east of Lord Howe Island, with a flesh-footed shearwater. Photograph: Patrick Kavanagh.

LORD HOWE ISLAND • 2001

The Tree Lobsters of Ball's Pyramid

For eighty years the rarest insect in the world survived on the side of a volcanic spike in the Tasman Sea, apparently all of it under a single bush.

Lord Howe Island, a crescent of forest and reef about 600 kilometres off the coast of New South Wales, used to have an insect so common that people fished with it. Large, black, wingless stick insects, as long as a hand and heavy as a small bird, lived in the hollows of its trees and came out at night to eat the leaves. The islanders called them tree lobsters and used them for bait.

In 1918 the supply ship *Makambo* ran aground on the island. Black rats came ashore. Within a couple of years there were no tree lobsters left. After 1920 nobody could find one, and

Dryococelus australis, the Lord Howe Island stick insect, was added to the long list of island species that had met a rat and lost.

Twenty kilometres to the south-east of Lord Howe stands Ball's Pyramid: the eroded plug of an old volcano, more than 550 metres of bare rock rising sheer out of rough water, only 300 metres across at its widest. It is the tallest sea stack in the world. Nobody lives there. There is almost nowhere to land. It was named for Lieutenant Henry Lidgbird Ball of the Royal Navy, who sighted it in 1788, and for most of the next two centuries it was looked at rather than visited.

Droppings

In 1964 a team of climbers from Sydney made an attempt on the summit and found, on the rock, the body of a large stick insect. Over the following years other climbers came across more fresh carcasses. Something seemed to be living out there. But expeditions to find live insects came back empty-handed.

In February 2001 two Australian scientists, David Priddel and Nicholas Carlile, went out to Ball's Pyramid with two companions to settle the question. They had done the arithmetic: there seemed to be just enough vegetation on the

stack to support a small population. They climbed about 120 metres up a steep grassy slope and found nothing but crickets. On the way down, in a crevice roughly 100 metres above the sea, they came to a single shrub, a *Melaleuca*, growing out of the rock. Underneath it were droppings. Large ones.

Stick insects of this kind feed at night. So Carlile came back after dark with a local ranger, Dean Hiscox, a camera and torches, and scrambled up the slope again. Under that one bush, in the leaf litter that had built up around its roots, they counted twenty-four Lord Howe Island stick insects. As far as anyone could tell, they were the entire wild population of the species: two dozen animals, under one bush, on the side of a spike of rock in the middle of the sea.

Adam and Eve

The authorities waited two years before taking any. A species that small can be wiped out by a single rockfall or a single dry season, but it can also be wiped out by people taking too many to save it. In 2003 a team from the New South Wales national parks service went back and collected two breeding pairs. One pair went to a private breeder in Sydney. The other, quickly nicknamed Adam and Eve, went to Melbourne Zoo.

There were difficulties at first. Then the insects started to breed, and kept on. They turned out to have a remarkable range of talents for a species on the brink. The young hatch bright green and active by day, and darken to a glossy black as they grow and switch to the night shift. Males and females form something like pair bonds, sleeping side by side with the male's legs draped over the female. And females can, if no males are available, produce young without them, which may be exactly how a handful of insects survived eighty years on a rock.

By 2008, when Jane Goodall visited, Melbourne Zoo had 700 insects and more than 11,000 eggs. By 2016 more than 13,000 eggs had hatched there, and eggs had been sent to Bristol, San Diego and Toronto to found insurance colonies. Budapest Zoo became the first outside Australia to breed them, in 2012; Prague joined the programme in 2025. A 2017 genetic study compared insects from Ball's Pyramid with museum specimens collected on Lord Howe a century earlier and confirmed what everyone had hoped: they were the same species, not a lookalike cousin.

There were surprises on the rock, too. In 2014 a team of climbers, there without permission, photographed live stick insects near the summit, around 500 metres up, in sedge growing from very thin soil. The population had never been as confined to its one shrub as it looked. It had simply been found in the one place anyone had searched.

Going home

The hard part, as ever, is not breeding the animal but undoing the reason it vanished. In 2019 Lord Howe Island carried out one of the most ambitious rodent eradications ever attempted on an

Twenty-four insects, under one bush, on the side of a spike of rock in the middle of the sea.



Lord Howe Island, where rats arrived in 1928 and the tree lobsters vanished. Photograph: Jocelyn.

inhabited island. Then, in April 2021, a rat turned up. A small remnant had evaded the poison; it was tracked down, the monitoring was changed, and the work goes on. Plans have been drawn up to test a return on Blackburn Island, an islet in the Lord Howe lagoon, before any release on the main island.

So, at the time of writing, the tree lobsters have not yet gone home. Thousands of them are living in zoos on three continents, descended from a handful of insects taken from under one bush. Somewhere on a spike of rock in the Tasman Sea, their wild relatives are still doing what they have done since 1920: eating, sleeping in pairs, and waiting for the rats to leave. ■



Ball's Pyramid from a boat at its base, 1965, the year it was first climbed. Photograph: John Game.

Dryococelus australis

Size	Up to about 20 cm long; wingless
Young	Hatch green and active by day; turn black and nocturnal as they grow
Trick	Females can reproduce without males



Wollemi pine foliage and a young cone. Photograph: John Tann.

NEW SOUTH WALES • 1994

The Tree in the Canyon

A parks officer, out exploring canyons with friends, walked past a tree that wasn't supposed to exist.

David Noble was a field officer with the New South Wales parks service, and in his spare time he went canyoning: abseiling into the narrow, wet gorges that cut through the sandstone of the Wollemi wilderness, north-west of Sydney, looking for ones nobody had been down before. In September 1994, with two friends, Michael Casteleyn and Tony Zimmerman, he dropped into a gorge so deep and sheltered that it held a patch of rainforest at the bottom. Among the trees was one he didn't recognise.

It was tall, with fern-like foliage and some of the strangest bark he had ever seen: dark brown, knobbly, bubbled all over. People have since compared it, not entirely unkindly, to a bowl of Coco Pops. Noble knew enough botany to know that he didn't know what it was, and he took a sample.

Botanists at the parks service and the Royal Botanic Gardens in Sydney worked it out. It was a new genus in the Araucariaceae, the ancient family of conifers that includes the monkey puzzle, the Norfolk Island pine and, most closely of all, New Zealand's kauri. It belonged to a lineage that had otherwise been known, if at all, only as fossils. It was named

Wollemia nobilis: for the wilderness, and for the man.

A family of one

It is the only species in its genus, and there are fewer than sixty adult trees in the wild, in a few stands not far apart. Many of them have multiple trunks, some of them dozens, sprouting from the same base, which makes the trees hard to count and suggests they are very old. Every wild Wollemi pine that has been tested is genetically almost indistinguishable from every other. At some point, thousands of years ago, the species passed through a bottleneck so narrow that its whole surviving population is, to all intents, one family.

That makes it extraordinarily vulnerable, and the location of the grove has been kept secret ever since. It has not been secret enough. In 2005 some of the wild trees were found to be infected with *Phytophthora cinnamomi*, a root-rotting water mould, which park rangers believe was carried in by unauthorised visitors.

Then came fire. In the summer of 2019–20, when bushfires burned across south-eastern Australia, the blaze reached the Wollemi wilderness. Specialist firefighters were winched down into

the gorge to set up an irrigation system around the trees, and aircraft dropped fire retardant around them. The fire came through. The grove survived.

Hiding in plain sight

The strategy that may save the Wollemi pine in the long run is the opposite of secrecy. Botanists found early on that the tree could be grown from cuttings, and botanic gardens around the world now hold them. From the mid-2000s they were sold to the public, and Wollemi pines are now planted in parks and gardens, and grown in pots as Christmas trees. The idea is simple: if the tree is everywhere, people have less reason to go looking for the original grove, and the species no longer depends on one gorge.

It is an odd fate for a tree whose lineage outlived the dinosaurs. The last of its line, found by accident at the bottom of a canyon in 1994, now sits on patios. ■



Morepork and laughing owl, J. G. Keulemans, 1888.



North and South Island piopio, J. G. Keulemans, 1888.

GALLERY

The Plates

In 1888 Walter Buller, a Wellington lawyer and New Zealand's most famous ornithologist, published the second edition of *A History of the Birds of New Zealand*, illustrated with hand-coloured lithographs by the Dutch artist J. G. Keulemans, who lived and worked in England. They are some of the most beautiful bird pictures ever printed, and several images in this issue, including the cover, come from them.

They are also a record of a catastrophe, made by people who were part of it. Victorian collectors widely assumed that New Zealand's birds were doomed to be

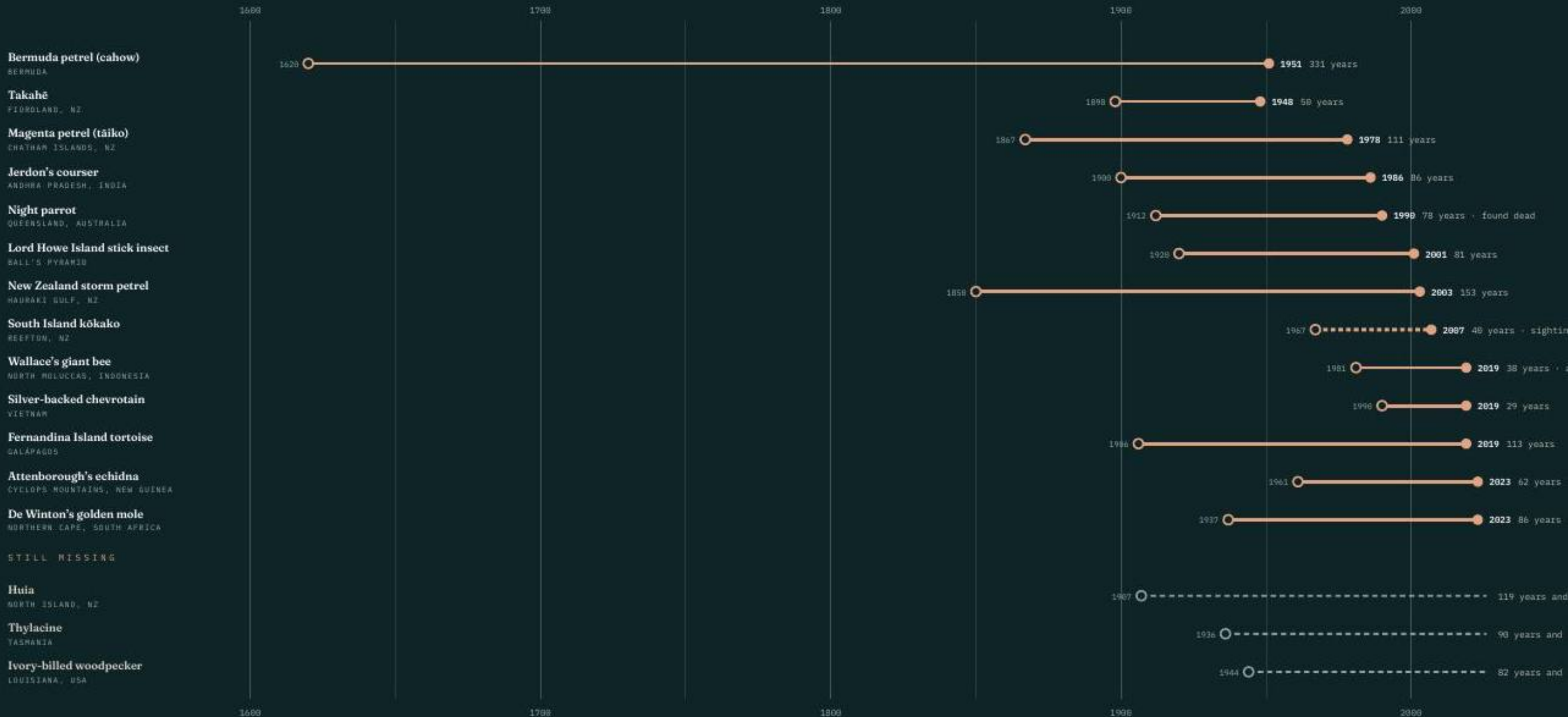
replaced by "more vigorous" European species, and concluded that the sensible thing was to get good specimens before they went. Buller himself collected eighteen huia on one expedition in 1883. In his *History* he describes a pair of huia "caressing each other with their beautiful bills" in a tree overhead, until "a charge of No. 6 brought them both to the ground together". He felt, he wrote, "almost glad that the shot was not mine, although by no means loth to appropriate 2 fine specimens".

Of the birds on these two pages, the morepork, on the left of the opposite page, is still calling in

the bush at night. Its larger cousin, the whēkau or laughing owl, was gone by about 1914. Both piopio, the North and South Island thrushes, were last confirmed around the turn of the twentieth century. The pictures outlived them. ■

Lost, and found again

Each line runs from the last record of a species to the day someone found it again. Hollow circle: last seen. Solid: found. Dashed lines at the bottom are still running.



66,000,000

Coelacanth

Absent from the fossil record since the Cretaceous. Caught off South Africa, 1938.

11,000,000

Laotian rock rat

Its family was known only from fossils. Found in a Laotian market and described in 2005.

70

Mountain pygmy possum

Described from fossil bones in 1896. Found alive at a ski resort on Mount Harthorn, 1966.

45

Chacoan peccary

Described from fossils in 1930. Recognised alive in the Paraguayan Chaco in the 1970s.



A Bermuda petrel, or cahow, over the Atlantic. Photograph: Festive Coquette.

BERMUDA • 1951

Night Voices of the Isle of Devils

The cahow was eaten to oblivion by Bermuda's first colonists and missing for three hundred years. A fifteen-year-old boy was there when it was found, and then he spent his life bringing it back.

Spanish sailors in the sixteenth century regarded Bermuda with suspicion. The reefs were bad enough, but the islands were also loud. At night, the story goes, the air above them was full of wailing and moaning, and they became known as the Isle of Devils. Bermudians will tell you the devils were birds: a gadfly petrel the size of a pigeon that nested in burrows on the islands in huge numbers and called to its mate in the dark with a cry that sounded, to English ears, like *cahow*.

The Spaniards did learn to stop and eat them. In 1603 one Spanish captain, Diego Ramírez, wrote that his men could take up to four thousand birds in a night. The hogs they released on the islands to provide fresh meat for later voyages rooted the birds' burrows out of the ground.

Then, in 1609, the English ship *Sea Venture*, bound for Virginia, was wrecked on the reef. It is the wreck that is generally thought to have given Shakespeare *The Tempest*. Most of the survivors eventually sailed on to Jamestown in two boats they built from the

wreckage, but Bermuda's permanent English settlement dates from them, and the settlers ate cahows too, the fattest birds and the eggs, especially in the lean winter months when the birds came ashore to breed. They brought rats, cats and dogs, and burned the forest.

In 1616 the governor issued a proclamation "against the spoyle and havocke of the Cohowes", one of the earliest conservation laws in the Americas. It made no difference. By the 1620s the cahow was gone. For the next three centuries it was a word in old documents, and any seabird reported off Bermuda that might have been one was assumed to be something else.

The bird in the lighthouse

In June 1935 an unfamiliar seabird flew into the lighthouse at St David's, at the eastern end of Bermuda, and was killed. The naturalist William Beebe, who was working on the islands, sent it to Robert Cushman Murphy, a seabird specialist at the American Museum of Natural History in New York. Murphy identified it as a cahow. It was a single dead bird,

and it could have come from anywhere. But if it had come from Bermuda, the cahow would have to be breeding somewhere nearby.

Murphy came to look for himself in 1951, with Louis Mowbray of the Bermuda Government Aquarium. They brought along a local boy who was mad about birds: David Wingate, fifteen years old. On 28 January 1951, on a cluster of tiny rocky islets in Castle Harbour, a sheltered bay at Bermuda's eastern end, they found cahows nesting in deep crevices in the rock. There were eighteen pairs. They were probably the whole species.

The boy who stayed

Most people who are present at a rediscovery go back to their lives. Wingate didn't. He went to Cornell University to study zoology and came home in 1958 to take on the cahow, and in 1966 he became Bermuda's first government conservation officer, a job he held until 2000.

The problems he faced were the kind that make conservationists go grey. The nesting islets were tiny, about a hectare between

them, and short of soil for burrows. Rats swam out to them. White-tailed tropicbirds, a native species bigger and more aggressive than the cahow, were evicting cahows from the few good crevices and sometimes killing their chicks. A cahow pair raises at most one chick a year, and a young bird spends years at sea before returning to breed, so the population could not bounce back from anything.

Wingate's answers were practical and patient. He poisoned the rats. He fitted wooden baffles over burrow entrances, cut so that a slim cahow could get in but a tropicbird could not. He built artificial burrows out of concrete. And, because the islets were simply too small, he began restoring Nonsuch Island next door, sixteen acres that centuries of misuse had turned into something close to desert. He replanted it with thousands of Bermuda cedars, palmettos and other native plants, so that the island might one day look as it had before people arrived, and the cahow might have somewhere to go.

Moving house

The idea was taken up by Wingate's successor, Jeremy Madeiros, who identified the biggest remaining threat as the islets themselves: low, eroding, and exposed to hurricanes and rising seas. In 2003 Hurricane Fabian destroyed about fifteen burrows outright and damaged most of the rest.

Seabirds tend to come back to breed where they fledged, so Madeiros, working with the Australian petrel specialist Nicholas Carlile, began moving chicks to artificial burrows on Nonsuch in 2004, about three weeks before they were due to leave for sea, and hand-feeding them squid and anchovies until they flew. Over five years 102



Cahow, *Pterodroma cahow*, at sea. Photograph: n88n88.

chicks fledged from Nonsuch. If Carlile's name is familiar, it should be: three years earlier, he had been one of the two scientists who climbed Ball's Pyramid and found the tree lobsters. The world of people who find lost things is a small one.

In February 2008 the first of the translocated birds came back to Nonsuch from the open Atlantic. In January 2009 one laid an egg there, the first cahow egg on Nonsuch for more than three hundred years.

Counting

In 1960 there were seventeen or eighteen breeding pairs, raising seven or eight chicks a year. In 2019 there were 132 pairs, fledging 72 chicks. In 2023 there were 164 pairs on six islands. Most of them nest in artificial burrows, and a camera in one of them streams the lives of a cahow pair to anyone

who wants to watch. The cahow is Bermuda's national bird, and the noise over the islands at night in winter, the noise that once frightened Spanish sailors, is gradually coming back. ■

THE LONG MIDDLE

Counting

Finding a species is the start. These are the counts that followed, for three birds in this issue. Dots are published counts; lines simply join them.

Takahe



Kākāpō



Cahow



Sources: NZ Department of Conservation (takahe); Kākāpō Recovery (kākāpō); Bermuda Department of Environment and Natural Resources (cahow pairs).



AOTEAROA - 1978 & 2003

Hiding at Sea

New Zealand storm petrels, Hauraki Gulf. Photograph:
Duncan (angrysunbird).

Two New Zealand seabirds, written off for more than a century, turned out to have been flying past us the whole time. Finding them meant going out at night, and going out to sea.

Seabirds are the easiest animals in the world to lose. They spend most of their lives far from land, they come ashore only to breed, often on remote islands, and many of the smaller petrels arrive and leave their burrows only in darkness. A species can be reduced to a few dozen pairs in a single steep gully and still pass, for decades, entirely unnoticed by people a few kilometres away. Both of these birds did exactly that.

I. The tāiko

On 22 July 1867 the Italian naval ship *Magenta*, crossing the South Pacific roughly midway between New Zealand and South America, collected a dark-backed, white-bellied petrel that no naturalist had seen before. It was named for the ship, *Pterodroma magentae*, the Magenta petrel, and for 111 years the specimen in Turin was the only one known.

It was not, in fact, a stranger to New Zealand. On Rēkohu, the main Chatham Island, eight hundred

kilometres east of Christchurch, bones in middens showed that the tāiko had once been the most abundant burrowing seabird on the island and an important food for Moriori, the islands' first people. Pigs, cats, rats and the clearing of the forest had all but wiped it out. Nobody could be sure that the bones on the Chathams and the skin in Italy belonged to the same bird.

The ornithologist David Crockett, from Whangārei, spent years trying to find out. Through the 1970s he led a series of expeditions to the south-west of Chatham Island, following up local reports of a strange bird. The method was hard, cold and nocturnal: standing on the hills in the dark with powerful spotlights, waiting for petrels to fly into the beam. On 1 January 1978, in the Tuku valley, he caught a tāiko.

Finding the bird was one thing. Finding where it nested took another decade, and radio transmitters fitted to birds caught in the spotlights. The burrows, when they were found, were long tunnels dug

under dense forest on land farmed by the Tuanui family. In 1983 Manuel and Evelyn Tuanui gave more than a thousand hectares of it to the Crown to protect the birds. It is now the Tuku Nature Reserve.

The tāiko remains one of the rarest seabirds on Earth: fewer than 200 birds, with only around twenty breeding burrows known. Since 2006 a predator-proof fence has enclosed a hilltop at Sweetwater, and chicks have been moved there by hand so that they will return to breed somewhere cats and rats can't reach. Crockett died in 2023. The bird he chased through the dark for a decade is still here.

II. The storm petrel

The New Zealand storm petrel is a tiny thing, about the size of a sparrow, dark above with a white rump and a streaked white belly, and it feeds by pattering across the sea surface on long legs, wings raised, as if walking on water. In the nineteenth century a handful of

specimens were collected in New Zealand waters and ended up in museums in England and France. After about 1850 there were no more. For a century and a half the species was known from those few skins, and was listed as extinct.

On 25 January 2003 two birders, Sav Saville and Brent Stephenson, were on a boat near the Mercury Islands off the Coromandel Peninsula when a small storm petrel came pattering past that didn't look like anything it should have been. Their photographs were inconclusive. On 17 November that year, out in the Hauraki Gulf looking for other storm petrels, Bob Flood and Bryan Thomas photographed and filmed between ten and twenty of them off Little Barrier and Great Barrier islands. There was no longer any doubt. A

bird thought extinct for 150 years was feeding within sight of Auckland, in one of the busiest stretches of recreational water in the country.

What nobody knew was where it bred. Over the next few years researchers caught birds at sea, fitted three with radio transmitters, and followed them. It took until February 2013 for a team led by Chris Gaskin and Matt Rayner to track the birds to their burrows: in the forest on Te Hauturu-o-Toi, Little Barrier Island. A year later the first New Zealand storm petrel egg was found, white with a fine dusting of pink at one end, 31 millimetres long.

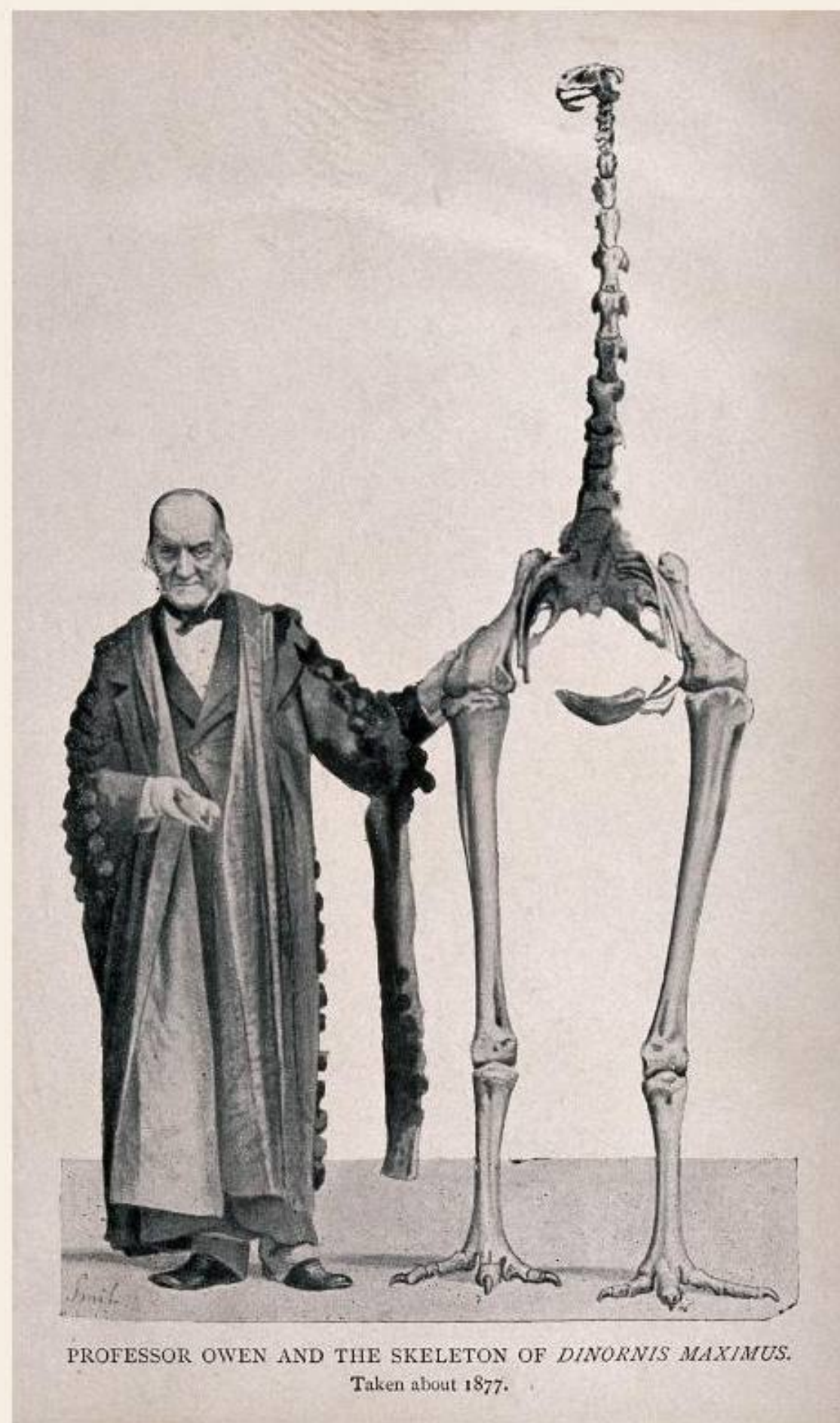
Hauturu has been one of New Zealand's most important island sanctuaries for more than a century, and its introduced cats and

Pacific rats have since been eradicated. One plausible explanation for the storm petrel's reappearance is that a remnant population had hung on there all along, too small to notice, and began to recover once the island's predators were gone. Nobody can know for certain. What is certain is that the birds are back on the water every summer, and that boatloads of birders now go out of Auckland to watch them walk on it. ■



"Walking on water": a New Zealand storm petrel pattering over the Hauraki Gulf. Photograph: Duncan (angrystunbird).

CHATHAM ISLAND TAIKO		NEW ZEALAND STORM PETREL	
Name	<i>Pterodroma magentae</i> , the Magenta petrel	Name	<i>Fregetta maoriana</i>
Size	About 38 cm long; around 450 g	Size	Sparrow-sized; feeds by pattering over the sea
Lost	1867 to 1978	Lost	About 1850 to 2003
Now	Fewer than 200 birds; about 20 known breeding burrows	Now	Breeds on Te Hauturu-o-Toi; first nest site found in 2013



PROFESSOR OWEN AND THE SKELETON OF *DINORNIS MAXIMUS*.
Taken about 1877.

Richard Owen and the skeleton of a giant moa, about 1877. Wellcome Collection.

ESSAY

Made, Not Found

Nobody is going to rediscover a moa. So a Texan company has offered to make one. Rediscovery and resurrection sound like the same miracle. They are not, and the difference matters.

There is no animal New Zealanders would rather find alive than a moa. There were nine species. The largest, the two species of *Dinornis*, could reach around 3.6 metres tall with the neck stretched up, and weighed about as much as a big stag. They were the only birds known to have lost their wings completely, not even a stub of bone remaining. Their closest living relatives, it turns out, are the tinamous of South America, small, dumpy birds that can still fly.

Polynesian voyagers reached Aotearoa some time before 1300. Within a century or so, according to radiocarbon dates from oven sites and middens, every moa was dead. By about 1445 they were gone, and so was Haast's eagle, the enormous raptor that had hunted them. It was one of the fastest extinctions of large

animals ever recorded, and the moa have been haunting the country ever since.

They have also been sighted. Nineteenth-century surveyors reported giant footprints; whalers and sealers told stories of monstrous birds on the coast. In 1959 an elderly woman named Alice McKenzie said she had seen a moa in Fiordland as a girl in the 1880s. In 1993 three people walking in the Craigieburn Range produced a blurred photograph of a large animal that interested the Department of Conservation for a while; it was generally judged to be a red deer.

People who want to believe sometimes invoke the takahē: gone for fifty years, and found. It is a fair point about the limits of searching, and a poor one about moa. The takahē is the size of a large chicken and left footprints that a determined man could

follow. A moa is the size of a small horse. To still be out there it would have to have lived unnoticed for five centuries in country crossed every week by hunters, trappers and deer-cullers, without leaving one convincing track, feather, dropping or bone. The moa is not missing. It is extinct. It will not be found.

The offer

On 8 July 2025, a company called Colossal Biosciences announced that it was going to bring it back anyway.

Colossal, based in Dallas, is the best known of the de-extinction companies, valued after its January 2025 funding round at more than ten billion US dollars. It has said it intends to produce woolly mammoth-like elephants, a thylacine, a dodo. Its moa project is a partnership with the Ngāi Tahu Research Centre at the

University of Canterbury, which is described as leading it, and with Canterbury Museum, whose collection of moa bones supplies the raw material. The film-maker Sir Peter Jackson, a moa enthusiast, and his partner Fran Walsh have put US\$15 million into it.

The plan, as Colossal has described it, goes roughly like this. Sequence the genomes of all nine moa species from DNA in bones. Work out which genes made a moa a moa, by comparing them with its living relatives; the emu and the tinamous are the leading candidates. Edit those changes into the cells of a living bird. Then turn the edited cells into an animal.

That last step is harder for birds than for mammals. A bird can't be cloned the way Dolly the sheep was, because a bird's egg can't be removed, emptied and refilled with a new nucleus. The likely route runs instead through germ cells, the precursors of sperm and eggs, edited and introduced into a host bird so that it lays eggs carrying the new genes. And no living bird lays an egg big enough for a giant moa. In May 2026 Colossal announced what it called a major breakthrough: an artificial egg, a transparent shell made of a silicone-based lattice that lets oxygen in, from which it had hatched twenty-six chickens. "There's currently no surrogate bird species that lays an egg big enough," its chief biology officer, Andrew Pask, told RNZ, "so we had to make the Colossal artificial egg."

What would hatch

It is worth being precise about what this could produce, because Colossal has already shown us.

In April 2025 the company announced the birth of three "dire wolves", Romulus, Remus and Khaleesi, white, heavy-set pups that appeared on magazine covers around the world. Dire wolves died out more than ten thousand years

ago. The pups were grey wolves whose genomes had been edited in about twenty places to give them some of the dire wolf's traits. The IUCN's specialist group on wild dogs said flatly that they were neither dire wolves nor proxies of dire wolves. A month later Colossal's own chief scientist, Beth Shapiro, described them as "grey wolves with 20 edits", and said that bringing back an animal identical to one that used to be alive is not possible.

A Colossal moa, then, would be a large flightless bird, probably built from an emu, with some moa genes, hatched from a plastic egg and raised by people. It would have no moa parents to learn from, and nothing to learn about: much of the forest it browsed has been felled or changed beyond recognition, the plants it ate have spent five hundred years evolving without it, and the eagle that hunted it is gone. The University of Otago ancient-DNA specialist Nic Rawlence, one of the project's most outspoken critics, has called it a genetically modified emu. Another Otago zoologist, Philip Seddon, said when it was announced that it had "nothing much to do with solving the global extinction crisis and more to do with generating fundraising media coverage". Colossal rejected that.

The case for

The project's supporters have real arguments, and they deserve a hearing.

Ancient genomes are scientifically valuable whatever happens next. They can tell us how many moa there were, how their populations rose and fell, how they were related, and what Aotearoa was like before people arrived. Tools developed for an impossible goal can turn out to be useful for possible ones. An artificial egg that lets you watch a chick develop, Pask argues, opens up new ways to

A Colossal moa would be a large flightless bird, hatched from a plastic egg and raised by people, with no moa parents to learn from.

help bird species that are struggling now. The money, too, is not being taken from the Department of Conservation; it is private money that would mostly never have gone to conservation at all. And the project, its partners say, is being steered by Māori, for whom moa are taonga, not by a lab in Texas.

The case against

The strongest objection is not that it can't be done. It is what the doing teaches us.

Every rediscovery story in this magazine has the same moral. The takahē, the cahow, the tree lobster, the tāiko: each was found alive because enough of it had survived, somewhere, by itself, carrying not only its genes but its knowledge, how to find food, how to choose a mate, where to dig a burrow, when to come home. Each was then saved, slowly, by people who changed the place it lived in: shot the deer, poisoned the rats, built the fences, planted the trees. None was saved by a machine.

De-extinction offers a seductive alternative story: that extinction is a technical problem, and that a sufficiently well-funded laboratory can reverse it. If that idea takes hold, it lowers the cost of losing



Moa bones at the Notoznis Expedition camp, Te Anau, c. 1949. The expedition that went to study the rediscovered takahē also dug in a moa hunters' cave. Archives New Zealand.

things. Why spend decades and millions keeping the last two hundred tāiko alive on one hillside, if we can always print a new one later? Colossal says it believes the opposite, that its technology is meant to stop extinctions as well as reverse them. But the stories a culture tells about extinction shape what it is willing to pay to prevent it.

And there is a question nobody can answer in a laboratory, which is who gets to decide. Moa were hunted by the ancestors of iwi throughout both islands. A taonga is not just an animal; it carries obligations. Colossal's position is that consultation is a matter for Māori to judge. That is respectful, and it is also, for a company with a ten-billion-dollar valuation, convenient.

Lazarus walked out

The magazine you are holding is named after a man who, in the story, was dead and buried for four days before he was called back. It is worth remembering the detail that gives the story its force: Lazarus walked out of the tomb himself. He was not assembled.

Lazarus walked out of the tomb himself. He was not assembled.



Moa leg bone, Dinornithidae. In 1839 Richard Owen was shown a fragment of a bone like this and concluded it came from a giant bird. Photograph: JC Harrison.



Black-footed ferret kits. Photograph: US Fish and Wildlife Service, Mountain-Prairie Region.

WYOMING • 1981

What the Dog Brought Home

The black-footed ferret was declared extinct in 1979. Two years later a ranch dog called Shep killed one. What happened next, all the way to a clone born in 2020, shows what de-extinction technology is actually good for.

1979 Declared extinct	1981 Shep's find, Meeteetse	1987 Last 18 wild ferrets taken into care	1991 First release, Shirley Basin	2020 Elizabeth Ann, a clone of Willa	2024 A clone has kits of her own
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The black-footed ferret is a slim, sand-coloured weasel with a black mask and black feet, and it lives almost entirely on prairie dogs. It sleeps in their burrows, hunts them at night in their burrows, and raises its kits in their burrows. Up to nine-tenths of its diet is prairie dog. So when the prairie dogs of the American West were poisoned by the million to clear land for cattle, and then struck by sylvatic plague, an introduced disease, the ferret went with them.

By 1974 the last known wild population, in South Dakota, had vanished. A few had been taken into captivity, but they bred badly, and by the end of the decade the last of them had died. In 1979 the species was declared extinct.

On 26 September 1981, on a ranch outside the small town of Meeteetse, in north-western Wyoming, John and Lucille Hogg found that their dog, Shep, had killed an animal they didn't recognise. Lucille took it to a local taxidermist. The taxidermist knew exactly what it was, and called the wildlife authorities.

A month later, biologists searching the prairie dog colonies around Meeteetse at night with spotlights found a living black-footed ferret, and then others. The extinct species had a population.

The second loss

It is the takahē story again, down to the dog. And like the takahē, the ferret immediately showed that finding a species and saving it are two different problems.

In 1985 plague struck the Meeteetse prairie dogs, and canine distemper, which is lethal to ferrets, broke out among the ferrets themselves. The population collapsed. Between 1985 and 1987 every remaining wild ferret that could be found was caught and taken into captivity, a decision that was bitterly controversial at

the time. The last was trapped in February 1987. There were eighteen of them: seven males and eleven females. Every black-footed ferret alive today is descended from seven of those animals that went on to breed.

The captive breeding worked. The first kits were born that same year at a research centre in Wyoming. In 1991 forty-nine ferrets were released into the Shirley Basin, in south-eastern Wyoming. Since then ferrets have been returned to prairie grasslands in eight US states, in Canada and in Mexico, and more than ten thousand kits have been born in human care.

Willa

The problem with descending from seven animals is that you are all cousins. Inbreeding leaves a species prone to disease and poor fertility, and there is no way to add new genes to a population that has none left to add.

Except, it turned out, from the freezer. In 1988 a female from the last wild group, named Willa, died, and cells from her body were frozen and stored. She had left no living descendants, so her genes were missing from the captive population. On 10 December 2020 a clone of Willa, named Elizabeth Ann, was born to a domestic ferret surrogate: the first clone of an endangered species native to the United States. Further clones followed, and in late 2024 one of them, Antonia, produced kits of her own. Willa's genes, absent for more than thirty years, were back in the breeding population.

Set this beside the moa. The same family of techniques that Colossal hopes will make a moa-like bird is already being used, quietly, for something more modest and more obviously useful: putting lost diversity back into a living species that is struggling. Nobody is claiming Elizabeth



A ferret leaves its carrier at a release site. Photograph: USGS.

Ann is a resurrection. She is a ferret, born into a species that still exists, because a ranch dog in Wyoming, forty-odd years ago, brought the right animal home. ■



Kōkako, North and South Island, J. G. Keulemans, 1898. The South Island kōkako, with orange wattles, has not been reliably seen since 1897.

BLUE-WATTLED CROW.
GLAUCOPIS WILSONI

(TWO THIRDS NATURAL SIZE)

ORANGE-WATTLED CROW.
GLAUCOPIS CINEREA.

STILL MISSING

The Ones That Haven't Come Back

For every takahē there are dozens of species that stayed lost. Four of them, and the people still looking.



Huia, female (left) and male, J. E. Keulemans, 1898.

The huia

Last confirmed: 28 December 1907, Tararua Range

The huia lived only in the forests of the southern North Island, and nothing else looked like it. Both sexes were glossy black with orange wattles and a white band across the tip of the tail, but the male had a short, stout, chisel-like bill and the female a long, slender, curved one, the most extreme difference between the sexes in bill shape of any known bird. The male hacked into rotten wood; the female probed the holes. They were said to be inseparable.

The huia was tapu, and its tail feathers were among the most

precious things a Māori chief could wear, traded the length of the country. Then Europeans wanted them too. Specimens were shot for museums and collectors; pairs were shot together. In 1901 the visiting Duke of York, later George V, was given a huia feather for his hatband at Rotorua, and the fashion took off. Rats, stoats and forest clearance did the rest.

On 28 December 1907 the naturalist W. W. Smith saw three huia in the Tararua Range. It is the last sighting anyone has accepted. There were credible reports into the 1920s, and a few after; nobody followed them up in time. In 1909 a search party went into the Ruahine Range looking for the bird

and came back with nothing. One of its guides, Henare Hāmana, could imitate the huia's call. Around 1949 he was taken into a Wellington studio and recorded whistling it. It is the only huia call that exists.

In 2024 a single huia tail feather sold at auction in Auckland for \$46,521.50, the most ever paid for a feather.

The South Island kōkako

Last accepted: 2007, near Reefton

The kōkako is a slate-grey forest bird with a black mask and a song often described as the most beautiful in New Zealand: slow, rich,

organ-like phrases sung in duet. The North Island species survives, rescued by predator control. The South Island species, with orange wattles where the northern bird has blue, has been on the edge of existence for most of a century.

Its record is a perfect illustration of how hard it is to declare something gone. The last authenticated sighting, for a long time, was at Mount Aspiring in 1967, and in 2007, forty years later, the Department of Conservation declared the species extinct. In 2013 the Ornithological Society accepted a sighting made by two people near Reefton in 2007, and the bird was moved from *extinct* to *data deficient*. The Department's own report was ambivalent: probably "functionally extinct", but "we are not convinced beyond reasonable doubt that the last individual of these taxa have died".

Reports keep coming: from the Heaphy Track, from the Old Ghost Road, from the bush near Tuatapere, where a hunter heard a strange call and saw a large grey bird on New Year's Day 2024. A feather found in 1995 and long argued over was retested in 2025 and confirmed, again, to belong to a blackbird. The South Island Kōkako Trust offers a reward of \$10,000 for evidence a panel of ornithologists will accept, and it is clear about what that means: a photograph or a feather, not a story.

The thylacine

Last known: 7 September 1936, Hobart

The thylacine, or Tasmanian tiger, was the largest carnivorous marsupial of modern times: a striped, dog-shaped animal with a stiff tail and jaws that opened astonishingly wide. It was gone from mainland Australia thousands of years ago, and in Tasmania it was hunted down under government and private bounties as a supposed

killer of sheep. The last known thylacine died at the Beaumaris Zoo in Hobart on 7 September 1936. In 1933 the naturalist David Fleay had filmed it in its cage for about forty-five seconds, and was bitten on the backside for his trouble. That grainy footage is the image most people carry of the species.

Under the rules of the time an animal could not be declared extinct until fifty years had passed without a confirmed record, and so the thylacine was declared extinct by the IUCN in 1982 and by the Tasmanian government in 1986. Sightings have never stopped. Neither have searches: camera traps, footprint casts, a two-year confidential hunt in the 1980s after a park ranger, Hans Naarding, reported a thylacine in his spotlight. Nothing confirmed. Australia marks National Threatened Species Day every 7 September.

The ivory-billed woodpecker

Last widely accepted: 1944, Louisiana

The ivory-bill was the giant of North American woodpeckers, black and white with a bone-coloured beak and, in the male, a flaming red crest, and it lived in the great old swamp forests of the American South until they were cut down. The last universally accepted sightings were in the 1940s. Then, in 2004, a kayaker reported one in the Big Woods of Arkansas, and the next year a team of scientists announced in the journal *Science* that the bird had been rediscovered, on the strength of a few seconds of blurry video. Thousands of people went looking. Nobody got a clear picture.

In 2021 the US Fish and Wildlife Service proposed taking the ivory-bill off the endangered list because it was extinct. The public response was so divided that the agency extended its deadline in 2022 and,



Thylacine skull. Photograph: JC Merriman.

in 2023, deferred the decision with no new date. At the last formal word, the ivory-billed woodpecker was neither officially alive nor officially extinct. It may be the only bird in the world whose status is, technically, pending. ■



Ivory-billed Woodpecker, Audubon
PIRELLA GÖTTSCHE LOWE

Ivory-billed woodpeckers, John James Audubon, *The Birds of America*.

Nearly

The kākāpō was never declared extinct. It came closer than any bird that survived.



Sirocco, the best-known kākāpō. Photograph: NZ Department of Conservation.

There is no other bird like the kākāpō: a moss-green, owl-faced, flightless parrot that weighs as much as a small cat, walks everywhere, climbs trees with its beak and feet, smells faintly of honey, and breeds mainly in the years when the rimu trees fruit heavily. In those years the males dig bowls into hillsides and spend whole nights inflating themselves like bagpipes to make a deep boom that carries for kilometres, so that females will walk across the island to inspect them.

All of this worked well for millions of years in a country with no land mammals but bats. Its defence against the eagles and harriers that hunted by daylight was to come out at night and, if threatened, freeze, trusting its camouflage. Against dogs, cats, rats and stoats, which hunt by smell, freezing is useless. By the early 1970s nobody could be sure the kākāpō still existed.

The Wildlife Service had been searching Fiordland since the 1950s. Between 1974 and 1978 it

found eighteen birds there, in steep glacial valleys. All of them were males. A species made entirely of males is a species with no future, and for a while it seemed that was what the kākāpō had become.

Then, in 1977, kākāpō were reported on Stewart Island/Rakiura, where nobody had thought to look. There were perhaps one or two hundred of them, and among them, crucially, were females. It was not a rediscovery of the species, quite, but it was a rediscovery of its future.

It was nearly lost again at once. Feral cats on Stewart Island were killing more than half the birds each year. Between 1982 and 1997 every kākāpō that could be caught was moved to predator-free islands: Little Barrier, Maud, Codfish Island/Whenua Hou, and later Anchor. In 1995, when the Kākāpō Recovery Programme was set up, there were 51 birds left in the world.

What followed is perhaps the most intensive care any wild

species has ever received. Every kākāpō has a name. Every one wears a transmitter. Their food is supplemented, their nests watched by camera, their eggs incubated when necessary and their chicks hand-raised when needed. Their sperm has been collected and artificial insemination attempted, and their genomes sequenced, every one, so that matings can be planned to protect what little genetic diversity is left.

It has been slow, and it has not always been happy: disease outbreaks, failed seasons, years without rimu fruit. But the numbers have gone the right way. By the latest count, in 2026, there were 325 kākāpō.

They are not safe. They may never be safe. Their entire world consists of a handful of offshore islands, and they will need people to look after them for as long as anyone can foresee. But almost every one of those 325 birds is alive because, in 1977, someone went somewhere nobody had bothered to look. ■

THE KAKAPO OR OWL PARROT.
STRINGOPS HABROPTILUS.
(ONE-HALF NATURAL SIZE)

Kākāpō, J. G. Meulmans, 1888.

How Lost Things Are Found Now

Geoffrey Orbell had a rifle, a theory and a good pair of boots. His successors have some new tools.

Dirt

De Winton's golden mole

, last seen 1937, found 2023

A blind, iridescent mole that “swims” through the sand dunes of South Africa’s west coast, leaving almost no trace on the surface. Nobody had recorded one for 86 years. In 2021 a team from the Endangered Wildlife Trust and the University of Pretoria walked the dunes collecting soil, and tested it for environmental DNA: the traces of skin and hair that every animal sheds into its surroundings. They also brought a sniffer dog trained to recognise golden moles. The DNA turned up in the soil from Port Nolloth to Lambert’s Bay. The mole was announced as rediscovered in 2023, and may be far more widespread than anyone had guessed.

Cameras

Attenborough's long-beaked echidna

, collected 1961, filmed 2023

A spiny, egg-laying mammal known to science from a single specimen collected in the Cyclops Mountains of New Guinea in 1961, named after Sir David Attenborough. In 2023 an expedition led by Oxford scientists, working with local Papuan partners, left trail cameras in the forest. When the memory cards were checked, one held the first footage of the animal alive.

Silver-backed chevrotain

, last recorded 1990, photographed 2019

A deer-like animal the size of a rabbit, unrecorded by science for nearly thirty years. Local rangers and villagers in southern Vietnam said they still saw it. Camera traps set where they pointed proved them right.

People who know

Wallace's giant bee

, rediscovered 1981, then lost again; filmed 2019

The largest bee in the world, a female with a wingspan of around six centimetres and jaws like a stag beetle's, living inside the nests of tree-dwelling termites in Indonesia's North Moluccas. In 2019 a single female was found, nesting in a termite colony in a tree on Halmahera, by the local naturalist Iswan Maujud, part of an expedition co-led by Clay Bolt and Eli Wyman.

Fernandina Island Galápagos tortoise

, last seen 1906, found 2019

Known from one male shot by a museum expedition in 1906, and from droppings and bite marks found over the following century. In February 2019 an expedition of park rangers and scientists found an elderly female. In 2021 genetic tests confirmed she belonged to the lost lineage.

Lists

In 2017 the conservation group now called Rewild published a list of the 25 “most wanted” lost species: animals and plants not recorded for at least a decade. Several on that list, including the golden mole, the echidna, the chevrotain and the giant bee, have since been found, and others have rotated in to replace them. A sister project, the Search for Lost Birds, launched a list in 2022 of about 160 bird species. By April 2026 it was down to 120. Five were re-found in 2025 alone, almost all of them photographed by birders and guides, local and visiting, who simply went and looked.

The lesson

Almost none of these animals were rediscovered by accident. They were found because someone decided they might still be there, chose a place, and went. Many were found because local people had never stopped seeing them, and someone finally listened. Lost, more often than we admit, means unlooked-for. ■



A camera trap on a tree trunk in the Cyclops Mountains of New Guinea. Photo: David Attenborough. Photo: David Attenborough. Photo: David Attenborough.

The Whistle

In 1909 a party went into the Ruahine Range, in the southern North Island, to look for huia. The bird had not been confirmed alive for two years, but there were rumours, and a hope of finding birds that could still be protected. One of the party's guides was Henare Hāmana, who also went by the name Harry Salmon. He knew the huia's calls, and could whistle them. Māori hunters had long drawn huia close by imitating them.

They found nothing.

Forty years later, in about 1949, a Moawhango farmer and amateur historian named Robert Batley took Hāmana to a recording studio in Wellington. Batley spoke a short introduction, describing a pair of huia feeding on a rotten tree. Then Hāmana whistled: a call, and an answer. The male, and then the female.

It is the only recording of a huia that exists. It is not a recording of a huia. It is a recording of a man remembering one, forty years after the last of them had gone quiet, imitating a sound he had once used to bring them close. You can hear it now, online. It is very short.

Every story in this magazine is about the hope that something we thought was gone might still be out there. It is a good hope, and sometimes it comes true. But the whistle is the other half of the truth. Most of the time, when something goes, what is left is a memory in one person, and then a recording of that memory, and then nothing at all.

The takahē, the kākāpō, the tāiko and the storm petrel are still here, and so are the people who spend their lives keeping them. Go and hear them while they can still answer. ■

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SEEN SOMETHING?

Most rediscoveries start with someone who reported what they saw.

In Aotearoa New Zealand, report sightings of threatened species to your local Department of Conservation office; for emergencies, such as wildlife being harmed, DOC's 24-hour line is **0800 DOC HOT** (0800 362 468). A photograph, a recording, a feather or a footprint, with the date and the place, is worth far more than a description. Note it, photograph it, and leave it where it is.

HUIA • *Heteralocha acutirostris*

*Last confirmed sighting:
28 December 1907.*

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